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Project Title: Exploring Game Reviews using N-gram and Part-Of-Speech Analysis

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Exploring Game Reviews using N-gram and Part-Of-Speech Analysis

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Abstract—The video game industry generates substantial revenue annually, making player feedback analysis crucial for developers and consumers. This study analyzes 21 video game reviews across three popular titles—Elden Ring, Stardew Valley, and Cyberpunk 2077—using N-gram and Part-of-Speech tagging methods to identify linguistic patterns that distinguish positive from negative sentiment. The dataset contains 12 positive and 9 negative reviews collected from online gaming platforms. N-gram analysis reveals that positive reviews emphasize quality descriptors such as "best game" and "amazing," while negative reviews focus on technical issues including "optimization issues" and "performance" problems. Part-of-Speech tagging shows similar grammatical structures across sentiments, with nouns and verbs dominating both categories, though positive reviews employ more descriptive adjectives. Unigram frequency analysis identifies "game" as the most common term in both sentiments, appearing 6 times in positive and 4 times in negative reviews. These findings enable game developers to identify specific praised features and common complaints, helping improve future releases and marketing strategies. The methods successfully extract actionable insights from unstructured review text, demonstrating practical applications in sentiment classification and feature extraction for the gaming industry.

Keywords—video game reviews, sentiment analysis, N-gram analysis, Part-of-Speech tagging, natural language processing, text mining

I. INTRODUCTION

In 2023, the global video game industry reached a market value in excess of \$180 billion, and player reviews significantly influence purchase decisions and game success [1]. With millions of players engaging in gaming across platforms such as Reddit, Steam, and Metacritic, online reviews have become an essential channel in allowing development teams to gauge interest, receive unfiltered feedback, and influence game sales. Beyond simple ratings, player reviews reveal nuance in how different players show their feelings towards a game. These expressions are different across genres – players might discuss mechanics in Elden Ring, creativity in Stardew Valley, or performance in Cyberpunk 2077. As the English Language is always evolving, understanding the language used in reviews offers a deeper view into the culture of gaming. These insights can highlight how language can be a bridge between gameplay experiences and player perception.

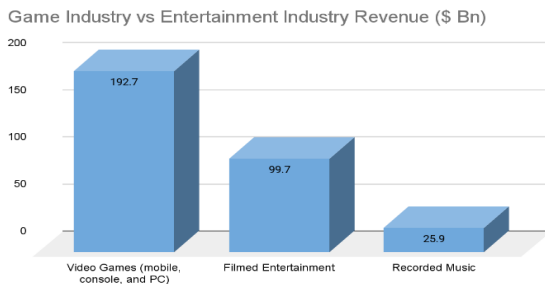


Figure 1: Graph showing the revenue figures for video games, filmed entertainment, and recorded music for 2021 [15]

With tens of millions of reviews readily available only, it is no surprise that together, one could create a text dataset that can be analyzed using natural language processing methods. Manually navigating and sorting reviews to reach observations, however, is not a productive use of developer or research time [2]. Natural language processing methods, in particular N-gram methods and Part-of-Speech (POS) tagging, are quick methods to detect sentiment string patterns and extract useful observations from player commentary [3]. N-gram procedures look at strings of words for identifying word clusters and word combinations to consider general linguistic patterns used in player feedback, while Part-of-Speech (POS) tagging provides a method to categorize words based on grammatical role to more effectively analyze meaning and detect sentiment [4]. These complementary approaches provide both lexical and syntactic insights into how players express opinions.

Using the reviews of three commercially successful games representing different genres: Elden Ring (action RPG), Stardew Valley (farming simulation), and Cyberpunk 2077 (open-world RPG), the study aims to answer the following research questions: What linguistic patterns distinguish positive from negative reviews? Which specific features do players praise or criticize most frequently? How can these insights benefit game developers and consumers? The findings contribute to understanding player sentiment in the gaming domain and demonstrate practical applications of NLP techniques for review analysis. This research provides actionable insights for game developers to improve their products, marketers to highlight strengths, and players to make informed purchasing decisions.

II. SCENARIO

A. Dataset Background

With a host of diverse opinions online about games, the dataset we selected for this study seeks to capture a representative snapshot of the online discourse. It includes reviews from three widely discussed games with distinct genres and player experiences:

- **Elden Ring:** A challenging action role-playing game known for its intricate mechanics and unforgiving difficulty
- **Stardew Valley:** A relaxing simulation game emphasizing creativity, farming, and open-ended gameplay loops.
- **Cyberpunk 2077:** An open-world game with expansive narratives and mind-blowing visuals, though initially criticized heavily at launch [12].

By selecting these titles, our analysis encompasses a variety of gameplay experiences and emotional reactions, allowing for a balanced foundation for deeper analysis.

B. Dataset Description

The dataset includes 21 video game reviews sourced from online gaming sites and review aggregation services. The reviews cover three popular games across different genres: Elden Ring (7 reviews), Stardew Valley (7 reviews), and Cyberpunk 2077 (7 reviews). Each game has both positive and negative reviews, providing balanced sentiment representation. The dataset includes the following attributes: - Game: The title of the video game being reviewed - review_text: The complete text of the user review - sentiment: Classification as either "Positive" or "Negative" - review_id: A unique identifier for each review. In the sentiment classification, there are 12 positive reviews and 9 negative reviews, which supports a comparison of properties in the relatively balanced data available. Reviews ranged from brief comments of a single sentence to longer evaluations of several sentences, with an average review length of roughly 14–15 words after preprocessing.

C. Data Collection Process

Reviews were chosen from Steam and Metacritic to represent an array of player views across several genres of games. Elden Ring represents challenging action RPGs with complex gameplay mechanics. Stardew Valley represents relaxing simulation games with creative freedom. Cyberpunk 2077 represents ambitious open-world games with both technical achievements and launch issues. The mixtures of both positive and negative reviews ensure that the analysis captures different linguistic patterns with distinctly different games and experiences to report.

D. Data Preprocessing

Text preprocessing was performed to clean and normalize the review data for analysis. The preprocessing pipeline included the following steps:

- 1) Lowercasing: All text was converted to lowercase to ensure consistency and prevent duplicate counting of the same words with different capitalizations.
- 2) Punctuation Removal: Special characters, punctuation marks, and numbers were removed using regular expressions, retaining only alphabetic characters and spaces. This step eliminates noise while preserving meaningful word content.
- 3) Tokenization: Review text was split into individual words (tokens) for subsequent analysis.
- 4) Stop word Removal: Common English stopwords such as "the," "is," "at," "which," and "on" were removed to focus on content-bearing words. The stop word list included articles, pronouns, prepositions, and auxiliary verbs that do not contribute to sentiment or topic understanding. However, the negation word "not" was retained, as it can reverse sentiment polarity.
- 5) Text Cleaning: The cleaned tokens were rejoined to create processed review text suitable for N-gram extraction and analysis.

This preprocessing approach follows standard NLP practices outlined in Natural Language Processing with Python [5] and ensures that subsequent analysis focuses on meaningful linguistic patterns rather than noise or common function words.

E. Dataset Statistics

After preprocessing, the positive reviews contained a total vocabulary of 85 unique words, while negative reviews contained 68 unique words. The word "game" emerged as the most frequent term in both sentiment categories, reflecting the domain focus. The preprocessed dataset provided a clean foundation for N-gram extraction and Part-of-Speech analysis to identify sentiment-distinguishing patterns.

III. RELATED METHODS

Natural Language Processing research have used n-gram and Part of Speech (POS) analysis for a long time to recognize linguistic and semantic patterns. The n-gram model identifies recurring word sequences that have been widely used to get contextual relationships and same tendencies in text based corpora. In practice, n-gram frequency analysis can be implemented using Python libraries such as NLTK or spaCy for tokenization and extraction. In recent studies, Nandwani et al. [7] emphasized that statistical language models such as n-gram remain effective in sentiment-related characteristic extraction, especially for speaking language like product or game reviews in these domains where oral language is frequent. Similarly, Islam et al. [8] emphasize that hybrid approaches combining n-gram frequency distributions with deep learning models have improved the detection of opinion polarity and intensity.

On the other hand, the part of speech (POS) tagging technique classifies tokens into grammatical species such as nouns, verbs and adjectives to support syntactic and semantic understanding. Common tools for POS tagging include NLTK, spaCy, and StanfordNLP, which provide reliable tokenization and grammatical labelling. Chiche and Yitagesu [5] provided a review of modern machine learning and deep learning methods for part of speech (POS) tagging, showing that these models are important in identifying affect-bearing adjectives in sentiment tasks. For example, identifying adjectives like amazing or boring in game reviews helps detect positive or negative tones. Verb patterns such as crashes frequently appear may mean functional complaints.

In the specific context of game review analysis, previous studies have explored how linguistic structures differ between genres and user attitudes. Guzsvinecz and Szucs [6] analyzed 2.5 million Steam reviews and discovered relations between review length, grammatical complexity, and sentiment trends across different game genres. Moreover, Viggiano et al. [9] identified the linguistic causes of sentiment in wrong classification, showed that negation, contrastive structures, and irony are major challenges in classifying game-related text. These findings support us to combine n-gram and Part of Speech (POS) methods for small scale linguistic exploration of game reviews.

Furthermore, other studies like A Comparative Study of Machine Learning Algorithms for Sentiment Analysis of Game Reviews [11] raise that using large-scale game review corpora for sentiment research, think that such community generated data react to real player experience. With these works and accessible corpora, such as the Steam and Metacritic reviews [13][14], they provide the methodological basics of our analysis. In our study, POS tagging and n-gram analysis were implemented using

Python with NLTK, ensuring reproducibility and direct applicability to our 21 selected user reviews.

We selected N-gram and Part-of-Speech (POS) analysis because they are effective for small sample sizes and provide complementary lexical and syntactic insights. N-gram methods focus on surface-level word patterns and frequent co-occurrences that reveal how players describe their experiences, while POS tagging uncovers the grammatical structure underlying those expressions. Together, these approaches enable a balanced view of both what players say and how they say it, which is especially useful when working with limited datasets such as short user reviews where deeper neural models may be impractical.

IV. TEXT EXPLORATION

For this section, we looked at the linguistic patterns we discovered across 21 user-generated for the 3 games: Elden Ring, Stardew Valley, and Cyberpunk 2077. Using Part-Of-Speech (POS) tagging and N-gram analysis, the goal was to uncover how different words and language were used to reflect a player's sentiment towards a game, as well as the distinctive characteristics of a game's genre.

A. Preprocessing

Before we could conduct the POS tagging and N-gram analysis, all the reviews were cleaned and standardized to ensure the structure of the reviews were consistent, and that there wasn't any noise that could affect the results. The cleaning processes were implemented in Python using both the **re** (regular expressions) and **NLTK** (Natural Language Toolkit). The steps we took are as follows:

1. Text normalization - the texts were converted into full lowercase for uniformity (e.g. 'Game' and 'game' are treated as the same word). All numbers, punctuation, and symbols were removed using a regular expression, which ensured that only the alphabetic characters remained for our linguistic analysis.
2. Stop word removal - Common English stop words such as *the*, *is*, *and*, *etc.* were removed using NLTK's built-in stop word list. However, we opted to intentionally retain the negation word 'not', as it is often used to reverse the sentiment of a given word e.g. while 'bad' has negative connotations, 'not bad' conveys positive sentiment.
3. Tokenization and Reconstruction - the cleaned text was then tokenized into words, and then rejoined together into continuous strings for the analysis.
4. Label cleaning - the 'sentiment' column was standardized by removing the extra spaces and converting all the labels into lowercase letters.

Below is the Python implementation of the preprocessing stage:

```

nltk.download('stopwords')
stop_words = set(stopwords.words('english')) # Add the stopwords into a set
df['Sentiment'] = df['Sentiment'].str.strip().str.lower() # tokenize the sentiment column and make everything lowercase
def clean_text(text): # Define a function to take care of the cleaning of the text
    text = re.sub('[^a-zA-Z]', '', text) # regex expression
    text = text.lower()
    tokens = [word for word in text.split() if word not in stop_words or word == 'not'] # explicitly keep the word 'not'
    return ' '.join(tokens)
df['clean_review'] = df['Review Text'].apply(clean_text)

```

Figure 2: Code for preprocessing stage

This process ensured that the data is free from any unnecessary elements while preserving the linguistic features. The resulting **clean_review** column contained the standardized text, which is optimized for use in the following stages, where we will perform linguistic pattern extraction using POS tagging and N-gram analysis.

B. N-Gram Analysis

N-gram analysis (unigrams, bigrams and trigrams) was conducted to identify the most recurring word patterns that characterize a game's discourse

1) Unigrams Analysis

Unigrams represent single words and help to identify the most frequently occurring terms in positive and negative reviews.

a) Top positive unigrams: The most recurring unigrams here include 'game', 'world', 'best', 'time', 'characters', and 'story'. These show the narrative immersion, world-building, and emotional satisfaction that these worlds provide - all of which are key aspects that are valued in open-world and role-playing games.

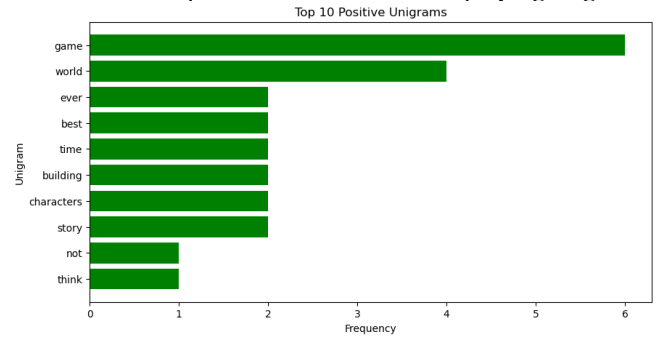


Figure 3: Top 10 most frequent positive unigrams

b) Top negative unigrams: The most recurring unigrams here are 'game', 'good', 'issues', 'not', 'feels', and 'got'. Surprisingly, there are words that appear here with positive connotations for negative reviews. However, it seems that, even in negative reviews, players acknowledge that the game has good characteristics, but express their disappointment with words like 'issues' and 'not'.

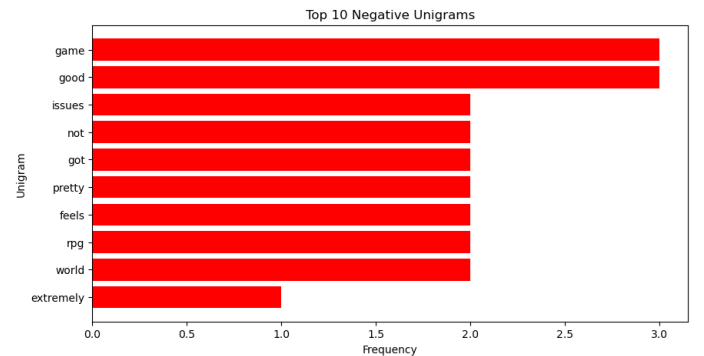


Figure 4: Top 10 most frequent negative unigrams

This analysis indicates that negative sentiment usually arises not from the story or gameplay, but rather technical flaws or performance limitations, which are commonly seen in large, graphically intense titles. For context, Cyberpunk 2077, one of the most pre-ordered and anticipated gaming titles of all time, launched disastrously. Now, the game has received hundreds of thousands of positive reviews, showing that the culprit of negative reviews are usually bugs. [12]

2) Bigrams Analysis

Bigrams (two-word combinations) provide insight into short phrases that convey tone and sentiment.

a) *Top positive bigrams:* Frequent pairs like “best game,” “overall best,” “game series,” and “feeling amazement” convey a sense of admiration and emotional connection. Phrases such as “best game series” indicate long-term loyalty, suggesting that players compare it favourably with other titles in the same franchise or genre.

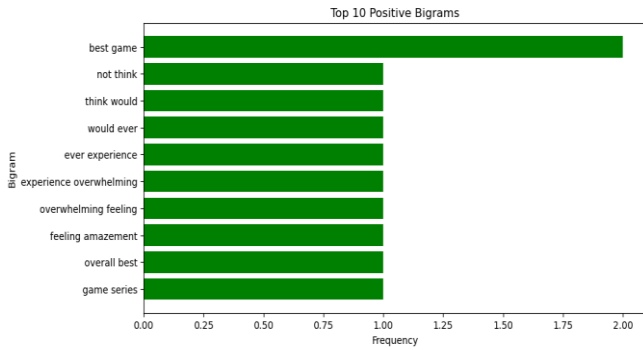


Figure 5: Top 10 most frequent positive bigrams

b) *Top Negative Bigrams:* Examples include “major optimization,” “pc major,” “game extremely,” and “extremely good.” These phrases show that while players may recognise the game’s strengths (“extremely good”), frustration with optimisation dominates.

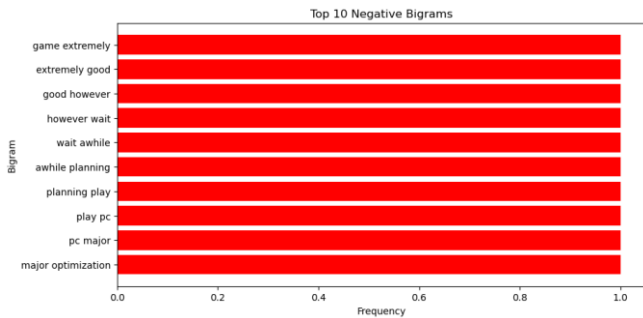


Figure 6: Top 10 most frequent negative bigrams

Bigrams clarify tone more precisely than unigrams. They reveal that positive players express enthusiasm and admiration, while negative players adopt a contrastive tone — often praising gameplay but lamenting performance flaws.

3) Trigrams Analysis

Trigrams (three-word sequences) capture slightly longer contexts and emotional nuance.

a) *Top Positive Trigrams:* Examples such as “ever experience overwhelming”, “experience overwhelming feeling,” and “overall best game” reveal emotional intensity. Players describe the game as a unique or awe-inspiring experience, indicating that the title succeeded in delivering deep emotional engagement — an important factor for role-playing or story-driven games.

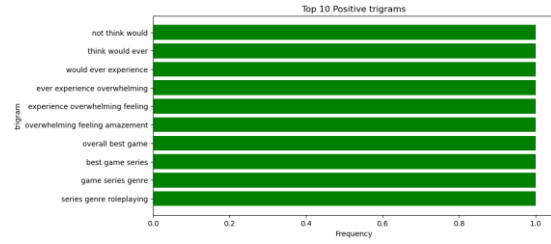


Figure 7: Chart showing top 10 most frequent positive trigrams

b) *Top Negative Trigrams:* Phrases like “major optimization issues,” “play pc major,” and “extremely good however” emphasize technical dissatisfaction and contrastive opinions. Even when acknowledging good aspects, players pivot quickly to criticism. This is typical of reviews where performance issues overshadow content quality.

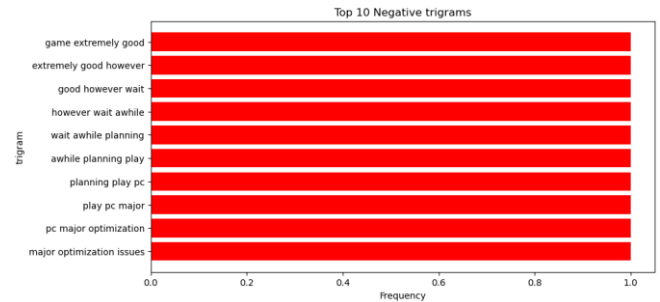


Figure 8: Top 10 most frequent Negative trigrams

Trigrams add emotional nuance, revealing how positive reviews emphasize astonishment, while negative reviews express dissatisfaction couched in contrastive phrasing (“good, however...”). These multi-word phrases reinforce the polarity already observed in simpler n-grams.

C. Part-Of-Speech (POS) analysis

To enhance the user accessibility of our POS analysis, we defined a dictionary of POS tags and their meanings. This dictionary can then be called when executing the POS analysis to provide a reference point. We also utilized this dictionary when printing the top positive and negative POS, ensuring that the reader is aware of these tags and their meanings. Below is the dictionary of tags and their meanings:

POS Tags Meanings

```

NN: Noun, singular or mass
NNS: Noun, plural
JJ: Adjective
JJR: Adjective, comparative
JJS: Adjective, superlative
RB: Adverb
VBG: Verb, gerund/present participle
VBD: Verb, past tense
VB: Verb, base form
VBP: Verb, non-3rd person singular present
VBZ: Verb, 3rd person singular present
IN: Preposition/subordinating conjunction

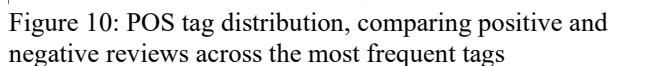
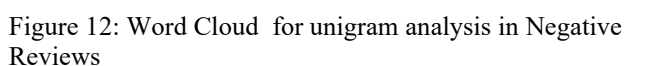
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Figure 9: POS tags dictionary

b) Negative Reviews

The negative texts similarly used many nouns (NN:23) and adjectives (JJ:18), but their tone differed. Adjectives like “repetitive,” “boring,” “empty,” and “dull” indicate dissatisfaction with the gameplay variety and environmental richness. The frequent use of verbs also shows that players actively describe actions going wrong or experiences that disappointed them.

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[illegible][illegible]

Across all the analyses conducted, there are consistent trends that can be seen between the positive and negative camps:

- Across both analyses, there is a clear split in how players communicate satisfaction versus disappointment. On the one hand, positive reviews rely heavily on descriptive adjectives and emotional language like 'overall best game'. This shows that there is an unquantifiable feeling of emotional fulfillment when playing these immersive games, and that this is the primary reward. Players use this language because they are describing a successful experience. Conversely, negative sentiment focuses more on measurable, technical issues. The N-grams explicitly showing 'major optimization issues' occurring frequently shows that players' frustrations are mostly triggered by system failures that interrupt the gaming experience, above all else. Ultimately, the type of language that is chosen by gamers acts as a key indicator of the game's perceived value. The language is more emotional for creative success, and functional for technical failure.

V. CONCLUSION

This study successfully applied N-gram and Part-of-Speech analysis to video game reviews, demonstrating that these NLP methods effectively identify linguistic patterns distinguishing positive from negative sentiment. The combination of lexical analysis through N-grams and syntactic analysis through POS tagging provided complementary insights into both what players say and how they express opinions.

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While limitations remain, particularly regarding dataset size and selection process, the methodology and findings provide a foundation for future work in gaming sentiment analysis. The successful application of established NLP techniques to the gaming domain demonstrates the value of computational text analysis for understanding player feedback in an increasingly data-rich industry. Ultimately, this study

highlights how NLP analysis can help game industry professionals better understand player sentiment by bridging linguistic research with practical application.

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